

Nishi Kanta Paul

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Research Profile

Computer Science graduate and **medical AI researcher** with hands-on experience in **multimodal vision-language models, biomedical image analysis, EEG modeling, explainable AI, and uncertainty-aware learning**. Experienced in building reproducible pipelines for radiology VQA, medical image segmentation, neurological prediction, and clinical decision-support research, with additional professional experience developing scalable production software systems. Interested in **trustworthy medical foundation models, multimodal learning, biomedical imaging and neural signal analysis**, with an emerging interest in **brain-inspired representation learning**.

Research Interests

Medical Foundation Models · Multimodal AI / Vision-Language Models · Trustworthy & Explainable AI · Biomedical Imaging & Neural Signal Modeling · Brain-Inspired Representation Learning

Education

B.Sc. in Computer Science and Engineering 2020 – 2024
Ahsanullah University of Science and Technology (AUST)
CGPA: **3.749 / 4.00** • Ranked **7th / 100 (Top 7%)**

Research Experience

Research Assistant — *Washington State University* Jun 2026 – Present

Project: Anomaly-Aware ForensiBlock

Supervised by Prof. Asma Jodeiri Akbarfam, School of Electrical Engineering and Computer Science

- Contributing to the **AI/ML and explainable AI pipeline** for a real-time authorized-user anomaly detection system.
- Developing **anomaly detection and XAI integration** for interpretable decision support and forensic traceability, under external faculty supervision.

Junior Research Assistant — *Elite Research Lab* Mar 2026 – Present

Project: BGD-SF PolySegNet — Boundary-Guided Polyp Segmentation

First Author

- Developing a **boundary-guided dynamic selective fusion network** for robust polyp segmentation in colonoscopy images.
- Designing multi-scale feature aggregation using **boundary cues and contextual semantic representations**, with cross-dataset validation, ablation studies, calibration analysis, and XAI evaluation.

Research Intern — *Elite Research Lab* Dec 2025 – Feb 2026

Project: MedVQA-X — Explainable Multimodal Medical VQA

- Designed and managed a **multimodal pipeline for ~18,000 radiology image-question pairs**, supporting reproducible training and evaluation of medical vision-language models.
- Developed and compared **LLaVA-Med, CLIP-ViT-L, and BioViL**-based approaches for radiology visual question answering.

- Integrated **GradCAM-based visual explanations** into the inference workflow and evaluated clinician-facing interpretability in a pilot setting.
- Performed large-scale preprocessing and annotation quality analysis, achieving **Cohen's $\kappa = 0.81$** for agreement.

Publications

Published

[C1] M. M. Kabir, M. A. Nawar, M. I. Remon, **N. K. Paul**, H. Bhuiyan, and M. M. Hoque, "Video-based Driver Distraction Detection using Transfer Learning," in *Proc. IEEE iCACCESS*, 2024. [🔗](#)

Accepted

[C2] **N. K. Paul** et al., "SleepEffFormer: Efficient CNN-Transformer with Transition-Aware Smoothing for Single-Channel EEG Sleep Stage Classification," in *Proc. 2026 IEEE Int. Conf. on Biomedical Engineering, Computer and Information Technology for Health (BECITH)*, 2026.

[C5] Shihabul, **N. K. Paul** et al., "WaveFoG: Wavelet-Gated Transformer for Parkinson's Freezing of Gait Detection Using Wearable Accelerometer Signals," in *Proc. IEEE BECITH*, 2026.

Under Review

[C3] Shihabul, **N. K. Paul** et al., "BioVerify: Evidence-Aware Biomedical Answer Verification with Confidence-Based Uncertainty Detection for Patient-Safe Healthcare NLP," in *Proc. 1st Int. Conf. on Next-Generation Electrical & Electronics, Computer Systems, and Technologies (iCONEECT)*, 2026.

[C4] **N. K. Paul** et al., "DR-LiteNet: A Lightweight Explainable Hybrid CNN Framework for Imbalanced Diabetic Retinopathy Grading via Adaptive SMOTE Fusion," in *Proc. iCONEECT*, 2026.

Industry Experience

Software Engineer I — *Brain Station-23 PLC*

Mar 2025 – Present

- Designed production-grade **data-intensive and event-driven systems**, including secure service communication, workflow processing, and monitoring infrastructure.
- Engineered an **eKYC data-processing pipeline** that reduced third-party integration latency by $\sim 40\%$, strengthening experience in reliable large-scale data workflows.
- Built secure multi-tenant services and audit pipelines using modern backend infrastructure, contributing practical experience relevant to **scalable and reproducible AI systems**.

Teaching Experience

Adjunct Lecturer — *International Standard University*

2026 – Present

- Teaching **Robotics, IoT, and Algorithms** to approximately 60 students and designing hands-on technical labs.

Selected Academic Honors

Scholarship for Outstanding Academic Performance

Fall 2020

Awarded after achieving **3.98 / 4.00** and ranking **1st / 100** in the inaugural semester at AUST.

Khondakar Ahmed Muktadir & Mosammat Khodeza Begum Scholarship

Fall 2022

Technical Skills

Medical / Multi-modal AI Modeling	PyTorch, TensorFlow, Transformers, CLIP, LLaVA, BioViL, GradCAM, SHAP CNNs, Vision Transformers, multimodal fusion, segmentation, uncertainty analysis, calibration
Data & Systems Programming	FastAPI, Django, PostgreSQL, MongoDB, Docker, Linux Python, C++, JavaScript, TypeScript

Additional

Competitive Programming: [ICPC Asia Dhaka Regional 2023 participant](#) · [Intra-AUST Programming Contest 2022 Champion](#) · [Codeforces Specialist \(rating 1421\)](#) · [500+ LeetCode problems solved](#).

References

Prof. Asma Jodeiri Akbarfam

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Research internship supervisor, MedVQA-X